

## LANGKAWI INTL, Malaysia

WMO: 486000

Lat: 6.3367N

Lon: 99.7264E

Elev: 9

StdP: 101.22

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 10            | 23.8       | 23.9 | 18.2                          | 13.1 | 29.0 | 19.2 | 13.9 | 28.7 | 6.4                   | 27.4 | 5.4 | 28.9 | 1.7                   | 60   | 0.363 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 3             | 7.2                    | 34.2            | 25.8 | 33.8 | 25.9 | 33.0 | 26.3 | 28.7                | 31.8 | 28.2 | 31.4 | 28.0 | 31.2 | 4.1                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.9                            | 24.0 | 31.5 | 27.2 | 23.0 | 30.9 | 27.1 | 22.9 | 30.8 | 93.3          | 31.8 | 91.0 | 32.1 | 89.9 | 31.6 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 6.2               | 5.3  | 4.6 | 22.2                       | 35.7 | 0.6                | 1.0 | 21.8                                               | 36.4 | 21.4       | 37.0 | 21.1       | 37.5 | 20.7       | 38.2 |    |    |
|                   |      |     | 20.0                       | 29.6 | 0.9                | 0.8 | 19.4                                               | 30.2 | 18.8       | 30.6 | 18.3       | 31.1 | 17.6       | 31.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 28.5   | 28.6  | 29.3  | 29.3  | 29.1  | 28.8  | 28.6  | 28.1  | 28.1  | 27.9  | 27.9  | 28.2  | 28.4  |
|                                                                   | DBStd       | 1.12   | 1.09  | 1.09  | 1.18  | 0.97  | 0.90  | 0.96  | 0.94  | 1.01  | 1.00  | 0.91  | 1.00  | 1.06  |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | CDD10.0     | 6762   | 578   | 539   | 599   | 573   | 582   | 558   | 562   | 561   | 536   | 556   | 547   | 571   |
|                                                                   | CDD18.3     | 3721   | 319   | 306   | 341   | 323   | 324   | 308   | 303   | 303   | 286   | 298   | 297   | 312   |
|                                                                   | CDH23.3     | 42956  | 3633  | 3702  | 4196  | 4030  | 3934  | 3652  | 3423  | 3411  | 3105  | 3181  | 3216  | 3474  |
|                                                                   | CDH26.7     | 16534  | 1411  | 1637  | 1871  | 1748  | 1583  | 1425  | 1220  | 1227  | 1032  | 1041  | 1114  | 1224  |
| Wind                                                              | WSAvg       | 1.9    | 2.6   | 2.7   | 2.2   | 1.8   | 1.6   | 1.7   | 1.6   | 1.7   | 1.7   | 1.6   | 1.8   | 2.4   |
| Precipitation                                                     | PrecAvg     | 2498   | 37    | 37    | 62    | 176   | 295   | 288   | 255   | 277   | 418   | 401   | 199   | 73    |
|                                                                   | PrecMax     | 3249   | 142   | 188   | 162   | 320   | 478   | 393   | 413   | 486   | 847   | 668   | 450   | 170   |
|                                                                   | PrecMin     | 1857   | 0     | 0     | 2     | 7     | 110   | 135   | 125   | 149   | 162   | 232   | 67    | 0     |
|                                                                   | PrecStd     | 359    | 44    | 50    | 48    | 88    | 116   | 85    | 98    | 107   | 183   | 133   | 98    | 50    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 34.1  | 35.2  | 35.8  | 34.2  | 33.2  | 33.1  | 32.2  | 32.2  | 32.2  | 33.0  | 33.2  | 33.2  |
|                                                                   |             | MCWB   | 25.7  | 25.1  | 25.7  | 27.2  | 27.9  | 27.9  | 27.3  | 27.1  | 27.3  | 26.7  | 26.7  | 26.1  |
|                                                                   | 2%          | DB     | 33.2  | 34.2  | 34.2  | 33.2  | 32.8  | 32.2  | 32.1  | 32.0  | 31.8  | 32.0  | 32.8  | 32.8  |
|                                                                   |             | MCWB   | 25.2  | 25.0  | 25.6  | 27.4  | 27.8  | 27.6  | 27.2  | 27.2  | 27.1  | 26.7  | 26.7  | 25.9  |
|                                                                   | 5%          | DB     | 32.8  | 33.2  | 33.2  | 32.9  | 32.1  | 32.0  | 31.7  | 31.2  | 31.1  | 31.2  | 32.0  | 32.0  |
|                                                                   |             | MCWB   | 25.1  | 25.1  | 26.2  | 27.4  | 27.5  | 27.5  | 27.2  | 27.0  | 27.0  | 26.8  | 26.6  | 25.9  |
|                                                                   | 10%         | DB     | 31.9  | 32.8  | 32.2  | 32.1  | 31.8  | 31.2  | 31.0  | 31.0  | 30.8  | 30.8  | 31.1  | 31.1  |
|                                                                   |             | MCWB   | 25.1  | 25.1  | 26.3  | 27.2  | 27.5  | 27.3  | 27.1  | 27.0  | 26.9  | 26.8  | 26.6  | 25.9  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.7  | 27.8  | 28.6  | 29.1  | 29.1  | 29.2  | 28.2  | 28.6  | 28.2  | 28.2  | 28.2  | 27.9  |
|                                                                   |             | MCDB   | 31.7  | 32.2  | 32.7  | 32.6  | 32.1  | 31.7  | 31.1  | 31.3  | 31.3  | 31.0  | 31.5  | 31.5  |
|                                                                   | 2%          | WB     | 26.9  | 27.2  | 27.9  | 28.5  | 28.5  | 28.5  | 28.0  | 28.0  | 27.7  | 27.8  | 27.6  | 27.2  |
|                                                                   |             | MCDB   | 31.0  | 31.6  | 31.9  | 32.0  | 31.5  | 31.3  | 31.0  | 30.8  | 30.7  | 30.7  | 31.0  | 31.0  |
|                                                                   | 5%          | WB     | 26.4  | 26.8  | 27.4  | 28.1  | 28.1  | 28.0  | 27.5  | 27.4  | 27.2  | 27.2  | 27.1  | 26.7  |
|                                                                   |             | MCDB   | 30.5  | 31.2  | 31.4  | 31.5  | 31.1  | 31.0  | 30.6  | 30.5  | 30.2  | 30.2  | 30.6  | 30.5  |
|                                                                   | 10%         | WB     | 26.0  | 26.4  | 27.0  | 27.7  | 27.8  | 27.6  | 27.1  | 27.0  | 26.9  | 26.9  | 26.7  | 26.3  |
|                                                                   |             | MCDB   | 29.9  | 30.9  | 31.0  | 31.0  | 30.7  | 30.6  | 30.3  | 30.2  | 29.8  | 29.8  | 30.1  | 30.0  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.7   | 7.6   | 7.2   | 6.2   | 5.7   | 5.8   | 5.7   | 5.6   | 5.6   | 5.7   | 6.1   | 5.9   |
|                                                                   |             | MCDBR  | 7.6   | 8.2   | 8.1   | 7.1   | 6.1   | 6.1   | 6.3   | 6.0   | 6.0   | 6.3   | 7.0   | 6.8   |
|                                                                   |             | MCWBR  | 3.1   | 3.5   | 3.5   | 3.3   | 3.2   | 3.3   | 3.4   | 3.2   | 3.2   | 3.1   | 3.1   | 3.0   |
|                                                                   | 5% WB       | MCDBR  | 6.7   | 7.4   | 7.1   | 6.3   | 5.8   | 5.9   | 5.9   | 5.8   | 5.7   | 6.0   | 6.4   | 6.2   |
|                                                                   |             | MCWBR  | 3.5   | 3.8   | 3.7   | 3.4   | 3.3   | 3.5   | 3.5   | 3.4   | 3.3   | 3.3   | 3.2   | 3.2   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.446 | 0.465 | 0.474 | 0.474 | 0.453 | 0.462 | 0.465 | 0.465 | 0.456 | 0.453 | 0.446 | 0.445 |
|                                                                   | taud        |        | 2.376 | 2.291 | 2.297 | 2.307 | 2.390 | 2.382 | 2.359 | 2.343 | 2.366 | 2.382 | 2.395 | 2.394 |
|                                                                   | Ebn at Noon |        | 873   | 864   | 856   | 842   | 841   | 822   | 824   | 838   | 858   | 864   | 866   | 866   |
|                                                                   | Edh at Noon |        | 124   | 138   | 138   | 135   | 121   | 120   | 123   | 128   | 127   | 124   | 121   | 120   |
| All-Sky Solar Radiation                                           | RadAvg      |        | 5.30  | 6.11  | 6.02  | 5.75  | 5.04  | 4.92  | 4.82  | 4.87  | 4.71  | 4.71  | 4.67  | 4.66  |
|                                                                   | RadStd      |        | 0.37  | 0.33  | 0.51  | 0.29  | 0.32  | 0.28  | 0.25  | 0.22  | 0.22  | 0.38  | 0.31  | 0.40  |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) | Regional (12 neighbors) | +0.22 | +0.14   | N/A    | +0.30 | +0.23   | +0.19 | N/A | N/A         | +94     | +94     |         |

Nomenclature: See separate page